Project 1 ChatGPT use:

We used ChatGPT to clean our data and create our differenced variables. Attached below is the prompt we submitted to a new chat window:

I have uploaded an Excel dataset containing UFC fight data. I need you to clean and transform this dataset for statistical/modeling analysis.

The dataset contains one row per fight and separate statistics for Fighter 1 and Fighter 2. I want you to create new **differential columns** for every numerical statistic that has a matching Fighter 1 and Fighter 2 version.

Primary transformation rule

For each matching numerical statistic, calculate:

Differential = Fighter 1 value − Fighter 2 value

This direction must be consistent throughout the dataset.

For example:

- fighter1_Significant strikes landed per minute − fighter2_Significant strikes landed per minute
- If Fighter 1 = 5.24 and Fighter 2 = 3.88, the new value should be **+1.36**
- If Fighter 1 = 3.03 and Fighter 2 = 4.75, the new value should be **−1.72**

Therefore:

- Positive = Fighter 1 has the higher value
- Negative = Fighter 2 has the higher value
- Zero = both fighters have the same value

Create differential columns for these matching pairs

1. fighter1_Reach − fighter2_Reach
2. fighter1_Significant strikes landed per minute − fighter2_Significant strikes landed per minute
3. fighter1_Strike Acc − fighter2_Strike Acc
4. fighter1_Strikes absorbed per muniet − fighter2_Strikes absorbed per muniet
5. fighter1_Striking Defense − fighter2_Striking Defense
6. fighter1_Takedown average − fighter2_Takedown average
7. fighter1_Takedown Accuracy − fighter2_Takedown Accuracy
8. fighter1_Takedown Defence − fighter2_Takedown Defence
9. fighter1_Submission average − fighter2_Submission average
10. fighter1_Wins − fighter2_Wins
11. fighter1_Losses − fighter2_Losses
12. fighter1_Draws − fighter2_Draws
13. fighter1_Height_in − fighter2_Height_in
14. fighter1_Age − fighter2_Age

Use clear column names such as:

- Reach_Diff
- Significant_Strikes_Landed_Per_Minute_Diff
- Strike_Accuracy_Diff
- Strikes_Absorbed_Per_Minute_Diff
- Striking_Defense_Diff
- Takedown_Average_Diff
- Takedown_Accuracy_Diff

- Takedown_Defence_Diff
- Submission_Average_Diff
- Wins_Diff
- Losses_Diff
- Draws_Diff
- Height_Diff
- Age_Diff

Important rules

Do **not** calculate a difference for `fighter1_Stance` and `fighter2_Stance` because these are categorical variables.

Do not alter `fight_outcome`. It is the target/outcome variable and must remain associated with the correct original fight.

Do not accidentally swap Fighter 1 and Fighter 2. Every differential must follow:

Fighter 1 − Fighter 2

Do not overwrite the original columns during the initial transformation. Keep the original Fighter 1 and Fighter 2 columns and append the new differential columns so the calculations can be verified.

Before creating the differential columns:

- Check that the paired numerical columns are actually numeric.
- Identify missing, blank, malformed, or non-numeric values.
- Do not silently replace missing values with zero.
- Report how many missing values exist in each relevant column.
- Preserve rows unless there is a clear reason they cannot be used.
- Correct obvious column-name formatting issues if appropriate, such as the typo in `Strikes absorbed per muniet`, but do not change the meaning of any variables.

Validation

After creating the columns, verify the calculations on several randomly selected rows.

For each checked row, show:

Fighter 1 value | Fighter 2 value | Calculated differential

Confirm that:

Differential = Fighter 1 − Fighter 2

Also check that the number of fight rows remains the same unless you explicitly identify and explain a necessary removal.

Important interpretation note

These columns should technically be treated as **Fighter 1 relative differentials**, not automatically as “advantage” columns.

For most variables, such as reach, striking accuracy, takedown accuracy, striking defense, and height, a positive value can reasonably represent Fighter 1 having more of that characteristic.

However, some variables have the opposite interpretation. For example:

- A positive `Strikes_Absorbed_Per_Minute_Diff` means Fighter 1 absorbs MORE strikes, which may be a disadvantage.

- A positive `Losses_Diff` means Fighter 1 has MORE losses, which may be a disadvantage.
- Age does not automatically have a universally positive or negative interpretation.

For now, keep the mathematical calculation consistently as **Fighter 1 − Fighter 2**. Do not reverse individual variables unless I specifically ask you to create direction-adjusted "advantage" features later.

Final output

Return a new cleaned Excel file containing the transformed dataset.

Also provide a concise summary stating:

- Original number of rows
- Final number of rows
- Differential columns created
- Missing-value issues found
- Any cleaning changes made
- Any rows removed, if applicable, and exactly why
- Confirmation that `fight_outcome` remained unchanged and aligned with its original fight

Do not make additional modeling assumptions or remove potentially useful variables without explaining them first.

ChatGPT returned this output:

ompleted the cleaning and transformation. The original Fighter 1/Fighter 2 columns remain in place, with the 14 differential columns appended.

Download the cleaned UFC dataset

- **Original fight rows:** 6,855
- **Final fight rows:** 6,855
- **Rows removed:** 0
- **Differential columns created:** 14, all calculated strictly as **Fighter 1 − Fighter 2**
- **Missing/malformed values:** 0 across all 28 numerical source columns used
- **Imputation:** None; no missing values were replaced with zero
- **Cleaning change:** Corrected `Strikes absorbed per muniet` to `Strikes absorbed per minute` for both fighters
- **Stance:** Left categorical and no differential was created
- **`fight_outcome`:** Verified unchanged for all 6,855 fights and still aligned with the original rows

I randomly selected five fight rows and checked **all 14 differentials on each**, for 70 validation calculations total. All passed. Representative checks:

Excel row	Variable	Fighter 1	Fighter 2	Differential
353	Reach	71	70	**+1**
1305	Significant strikes/min	3.75	2.08	**+1.67**
3135	Significant strikes/min	2.47	4.32	**−1.85**

Excel row	Variable	Fighter 1	Fighter 2	Differential
5429	Reach	71	77	**−6**
5593	Age	24	33	**−9**

The new features remain **Fighter 1 relative differentials**, including variables such as losses, strikes absorbed, and age. I did not reverse any of them into directional "advantage" features.

I also surfaced the Google Drive connector if you want to work directly with Sheets/Drive files in future analyses.

The cleaned and validated Excel file is still ready here:

Download the transformed UFC dataset

Google Drive is connected, but the direct Drive import failed because the local file handoff expired after the workbook was generated. The Excel file itself is unaffected and fully complete.

Attached is a link to the ChatGPT data cleaning conversation:

https://chatgpt.com/share/6ab4007e-8f38-83ea-b022-681d54b2cc7e